

TTT4150

Inertial Navigation Systems

Exercise 10

Inertial frame (i-frame)

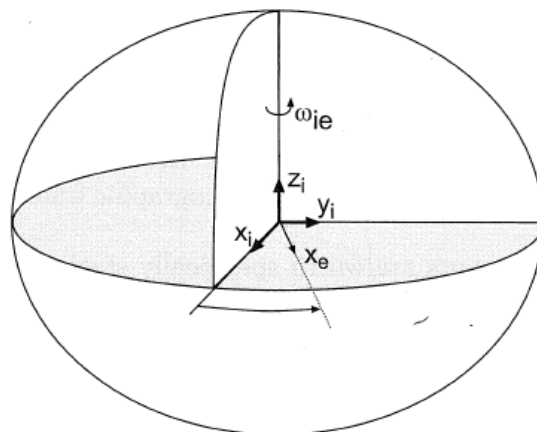
An inertial frame is a reference frame in which Newton's laws of motion apply. The origin of the system is arbitrary, and the coordinate axis may point in any three mutually perpendicular directions. *All* inertial sensors observe relative to an inertial frame.

It is often convenient to define an Earth Centered Inertial (ECI) frame which at a certain time has its origin coincident with the Earth's center of mass.

ECEF frame (e-frame)

The Earth Centered Earth Fixed frame has its origin fixed to the Earth's center of mass. The x-axis is extended through the intersection of the prime meridian and the equator, the z-axis coincides with the Earth's spin axis and the y-axis completes a right-handed orthogonal coordinate system.

Since the ECEF frame rotates with respect to inertial space it will *not* be an inertial frame.

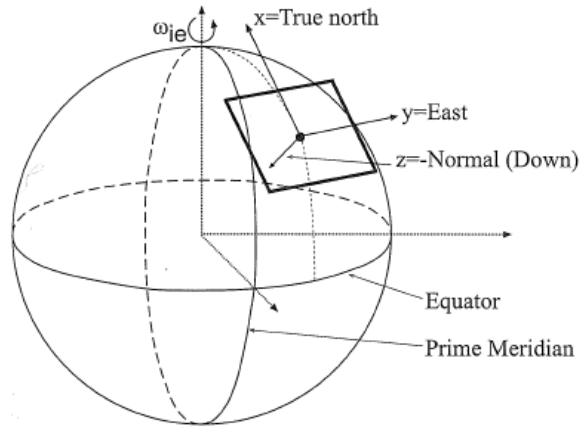


Geographic frame (g-frame)

The geographic frame is defined locally. This is a cartesian orthogonal coordinate system where the x-axis points towards north, the y-axis points towards east and the z-axis is aligned with the ellipsoidal normal pointing towards

the interior of the Earth (NED).

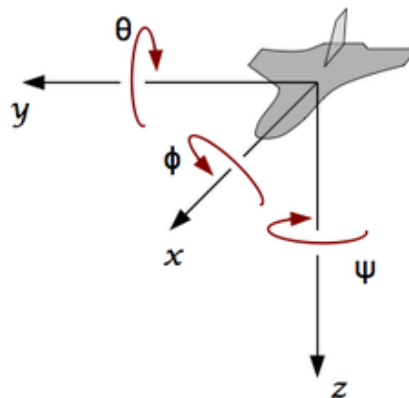
Since the origin of the geographic frame travels along with the vehicle, the axes of this frame rotates with respect to inertial space, thus geographic frame is *not* an inertial frame.



Body frame (b-frame)

In navigation applications, the objective is to determine the position and velocity of a vehicle based on various sensors attached to a platform on the vehicle itself. The body frame is rigidly attached to the vehicle of interest using its center of gravity as origo. The x-axis points in the forward direction of the vehicle, the z-axis points to the bottom of the vehicle and the y-axis completes the right-handed orthogonal coordinate system.

The three rotations around the x-axis, y-axis and z-axis are named *roll* (ϕ), *pitch* (θ) and *yaw* (ψ) respectively.



Inertial Navigation System (INS)

At some point in time an aircraft has the following position, speed and heading:

latitude	longitude	height	heading	speed
45°00'00"N	30°00'00"W	10 000 m	60°	800 km h ⁻¹

The body axes x^b and y^b are horizontal at this particular point in time.

1. Derive the matrix C_b^g that transforms vectors from b-frame into g-frame.

Hint: These angles are defined as Euler angles, and must be applied in the sequence $R_z(\psi) \rightarrow R_y(\theta) \rightarrow R_x(\phi)$.

2. Derive the matrix C_e^g that transforms vectors from e-frame into g-frame.

Now the aircraft enters a left turn, and over a period of 1 s the gyros in the onboard Inertial Motion Unit (IMU) is observing the following angular rates in the b-frame:

$gyro_{x^b}$	$gyro_{y^b}$	$gyro_{z^b}$
-0.1 ° s ⁻¹	0.1 ° s ⁻¹	-1.0 ° s ⁻¹

The rotations are defined *positive* counter clockwise as seen towards origo. The speed remains unchanged at 800 km h⁻¹ in x^b .

3. Update C_b^g at $t_k + \frac{1}{2}$ s.

Hint: To update C_b^g you will need ω_{gb}^b that can be computed from the gyro observations (ω_{ib}^b) together with the Earths rotation rate (ω_{ie}^e) as given in (3).

The Earths angular rate is $\omega_{ie}^e = 7292115 \cdot 10^{-11} \text{ rad s}^{-1}$. Assume the Earth being a sphere with $R = 6371 \text{ km}$

4. Use the updated C_b^g to compute the aircrafts position p_{eb}^g at $t_{k+1} = t_k + 1 \text{ s}$.

Over the same second as the gyros are observing angular rates, the accelerometers in the IMU are also observing accelerations in the b-frame.

acc_{x^b}	acc_{y^b}	acc_{z^b}
0 m s ⁻²	-3.9 m s ⁻²	-10.1 m s ⁻²

4. Compute the aircrafts velocity v_{eb}^g at $t_{k+1} = t_k + 1 \text{ s}$

Gravity is assumed to be 9.81 m s⁻² at sea level.

5. Compute the aircrafts position p_{eb}^g at $t_{k+1} = t_k + 1 \text{ s}$

Hint: Use the velocity v_{eb}^g

Note that in practice all these calculations will be done on a much higher rate than 1 Hz.

$$\dot{C}_b^g = C_b^g \cdot \Omega_{gb}^b \quad (1)$$

$$\Omega = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (2)$$

$$\omega_{gb}^b = \omega_{ib}^b - C_g^b \cdot \omega_{eg}^g - C_g^b \cdot C_e^g \cdot \omega_{ie}^e \quad (3)$$

$$\omega_{eg}^g = \begin{bmatrix} \frac{v_y}{(R+h)} \\ -\frac{v_x}{(R+h)} \\ -\frac{v_y}{(R+h)} \cdot \tan(\phi) \end{bmatrix} \quad (4)$$

$$a_{eb}^g = \hat{C}_b^g f_{ib}^b - (2\Omega_{ie}^g + \Omega_{eg}^g) v_{eb}^g + g_{eb}^g \quad (5)$$

$$C_b^g = \begin{bmatrix} c(\psi)c(\theta) & -s(\psi)c(\phi) + c(\psi)s(\theta)s(\phi) & s(\psi)s(\phi) + c(\psi)s(\theta)c(\phi) \\ s(\psi)c(\theta) & c(\psi)c(\phi) + s(\psi)s(\theta)s(\phi) & -c(\psi)s(\phi) + s(\psi)s(\theta)c(\phi) \\ -s(\theta) & c(\theta)s(\phi) & c(\theta)c(\phi) \end{bmatrix} \quad (6)$$

$$C_e^g = \begin{bmatrix} -s(\phi)c(\lambda) & -s(\phi)s(\lambda) & c(\phi) \\ -s(\lambda) & c(\lambda) & 0 \\ -c(\phi)c(\lambda) & -c(\phi)s(\lambda) & -s(\phi) \end{bmatrix} \quad (7)$$